



IIT ROORKEE



NPTEL ONLINE
CERTIFICATION COURSE

Human Resource Analytics

Lecture 01

Introduction to HR analytics

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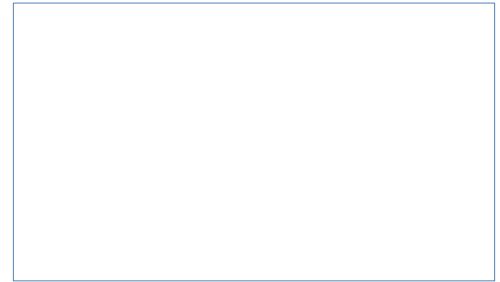
Faculty-Human Resource Management

XLRI-Jamshedpur



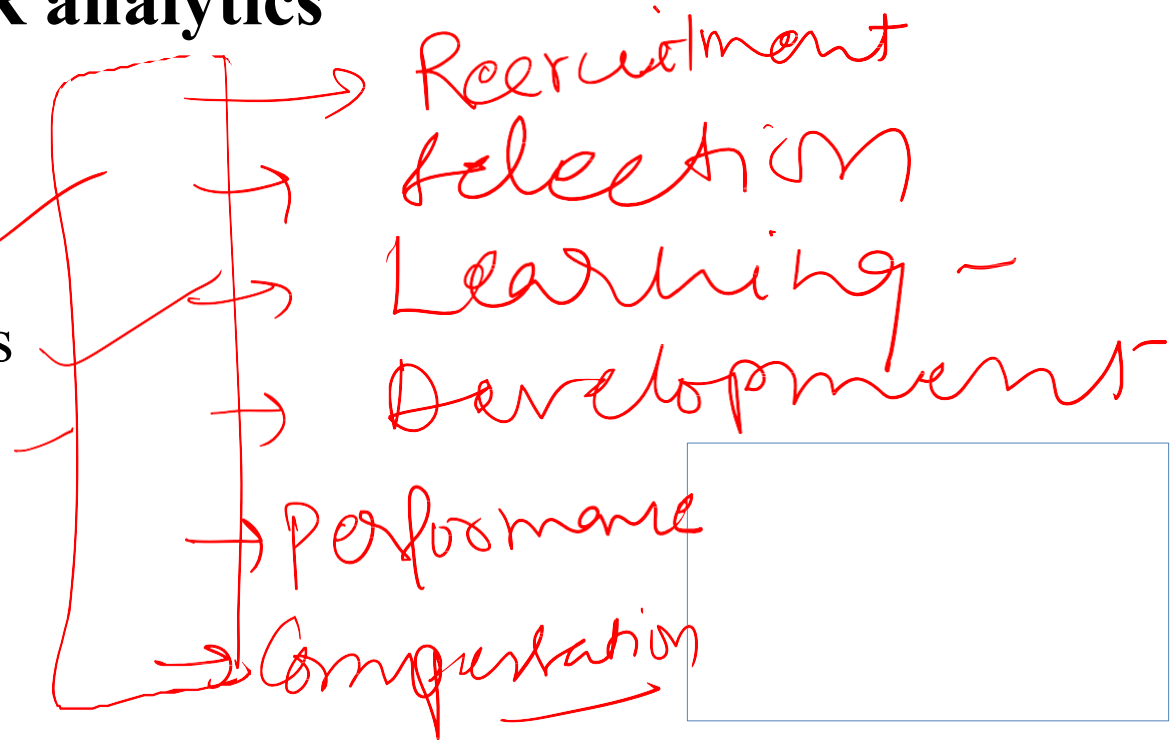
Content

- Introduction to HR analytics
- Questions that HR analytics answer
- Types of analytics
- Descriptive analytics tools
- Visualization perspective
- Predictive analytics tools
- Difference between MIS and Analytics



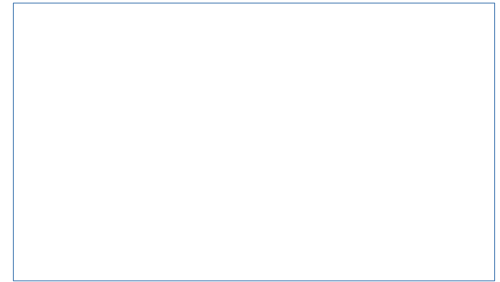
Introduction to HR analytics

- HR analytics
- People analytics
- Work force analytics



Questions that HR analytics answer

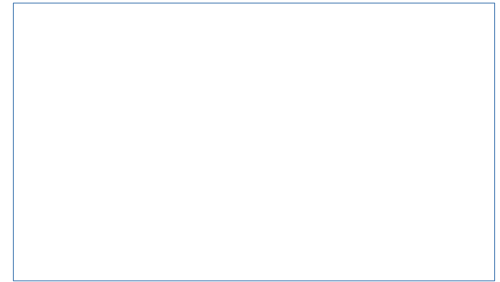
- What has happened? ✓
- What is happening?
- Where it is happening?
- How it is happening?
- What seems to be the underlying causes?



Types of analytics

- Descriptive analytics – What did happen
- Diagnostic analytics - Why did it happen
- Predictive analytics - What will happen
- Prescriptive analytics – What should happen or how can we make it happen

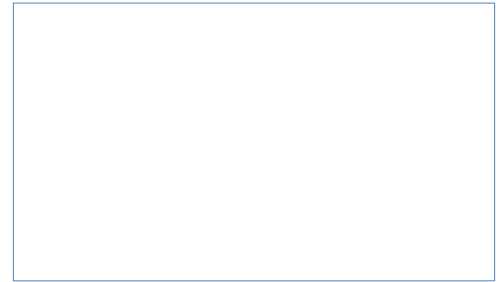
complexity → 200
optimi HR →



Descriptive analytics tools

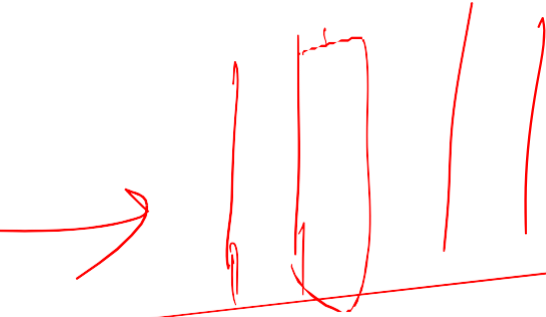
- Averages (Means)
- Standard deviation
- Variance
- Modes
- Median
- Counts
- Ratios

Average of your end
Average salary
found
male
0 25-30 -
30-35 -

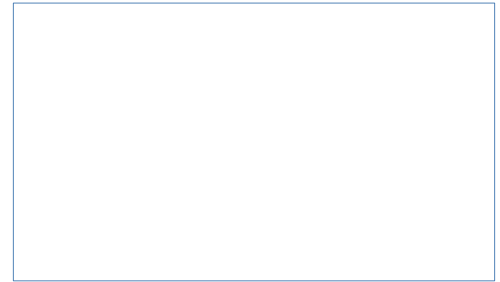
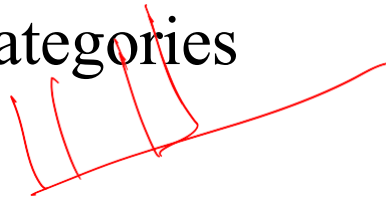


Visualization perspective ✓

- Histograms
- Showing proportion of things in pie charts
- Bar charts showing counts of things in different categories



Sources



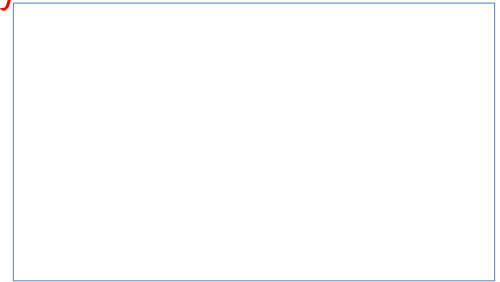
Diagnostic analytics

- Why did what we are observing occur
- Where did it occur?
- Is the metric we are monitoring (dependent variables) related in any way to the other things that we have collected data for (Independent variables)
- If so what is the strength of that relationship
- How much of the variability of our metric is accounted for from that data we have collected

— No. of Application

— No. of

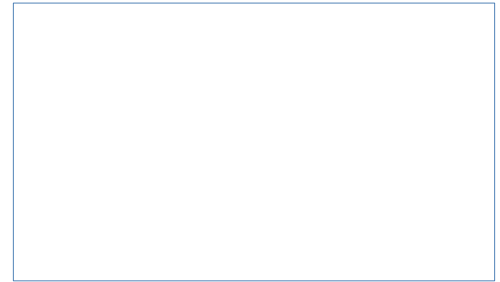
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Diagnostic analytics tool

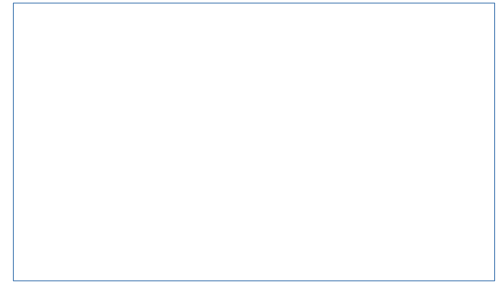
- Correlation and Regression ✓
- Analysis of variance ✓
- Factor analysis ✓
- Cross tabulation ✓
- Principal component Analysis ✓
- Correspondence Analysis ✓
- Multiple correspondence analysis ✓

ANOVA + t-test



Visualization perspective

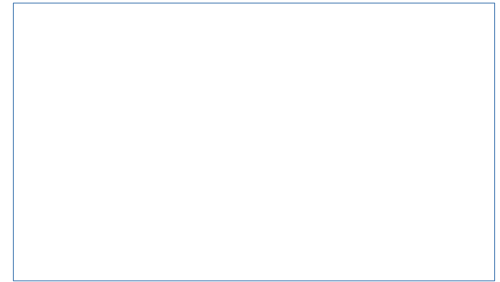
- Scatter plots ✓
- Regression plots ✓
- Plots of residuals ✓
- Box plots ✓
- Multiple density curves ✓



Predictive analytics tools

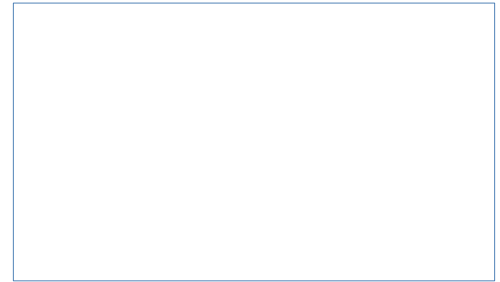
- Regression – Linear, curvilinear, logistic
- Decision trees and its variants
- Random forests

Discriminant analysis



Visualization perspective

- Line charts
- Scattergrams / correlation plots



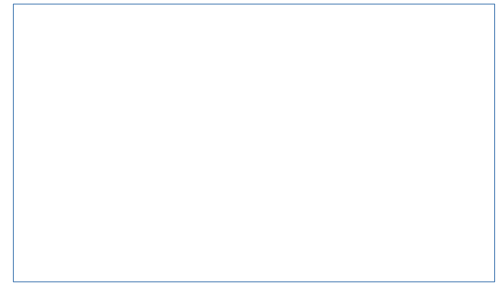
Prescriptive analytics

- Using the predictive model on new data either real data or vary the data in model to achieve a desired goals

Tools

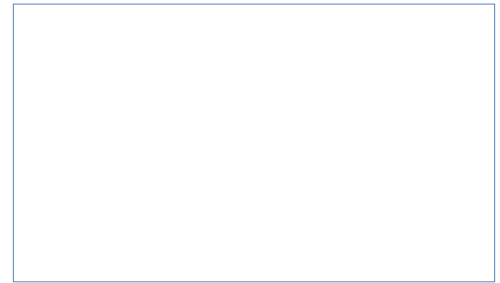
- Operation research (For example – optimization problems)

Mathematical models



Difference between MIS and Analytics

HRMIS is the source of the HR data. That may be used to make various types of metrics related to HR functions such as recruitment effectiveness metrics, training effectiveness metrics, performance metrics etc. HR analytics offers more than HR metrics through its potential to connect HR processes and decisions with organization performance, which is an avenue to elevating HRM to having a more strategic role and joining other business functions at the strategy table.



Basic Questions of HR analytics

- How does HR analytics work?

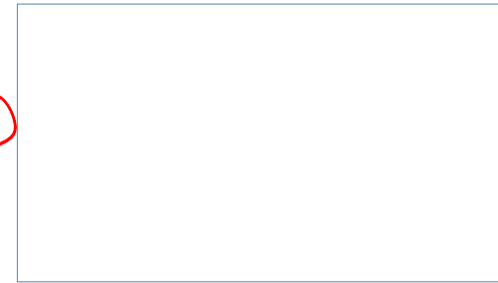
LAMP – Model

Logic, analytics, measures and processes

HR score card – Linking people, strategy and performance

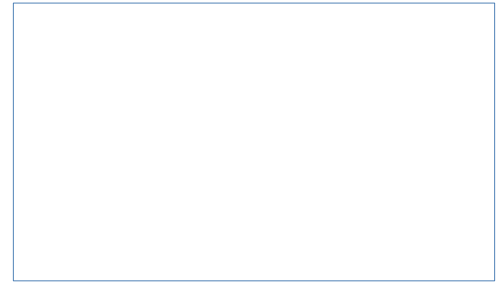
costs
Org. process
2 new

Q → List Variables → Data analysis



Basic Questions of HR analytics

- What does HR analytics procedure?



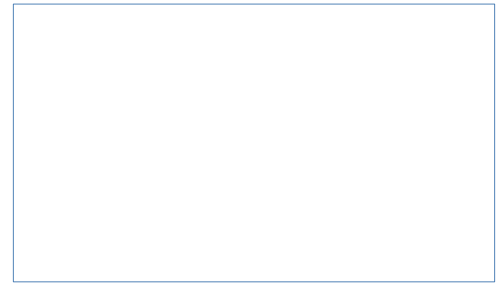
Basic Questions of HR analytics

- **What is required for HR analytics to succeed**

HR professional's analytical skills

HR information technology

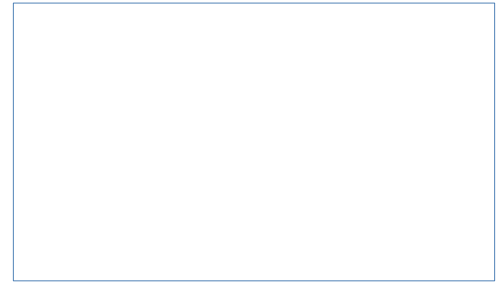
Knowledge of data analytics tools and techniques



Basic Questions of HR analytics

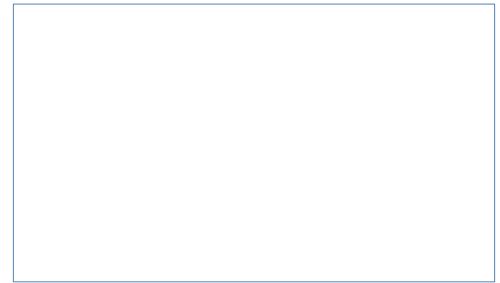
- **What are the outcomes of HR analytics**

HR analytics establish a link between HR processes, employee engagement, and store performance.



HR metrics are measures of key HRM outcomes, classified as efficiency, effectiveness or impact.

HR analytics is an evidence-based approach for making better decisions on the people side of the business, it consist of an array of tools and technologies, ranging from simple reporting of HR metrics all the way up to predictive modeling



Thank You

